OMAR REYAD SHOHIEB

AI-Engineer

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PROFESSIONAL SUMMARY

Computer Vision Specialist and AI student with hands-on experience designing, training, and benchmarking deep learning models for image classification, segmentation, and edge detection. Skilled in building end-to-end AI systems including CNN architectures, LLM-powered applications, and Retrieval-Augmented Generation (RAG) pipelines. Proficient in PyTorch, TensorFlow, OpenCV, and Scikit-learn, with strong foundations in NLP and recommendation systems. Currently deepening industry expertise through a Computer Vision training program at Fawry. Passionate about building production-grade vision systems backed by strong research and problem-solving skills.

TECHNICAL SKILLS

Programming Languages: Python, SQL Oracle, C++

Computer Vision: CNN Architectures, Image Classification, Segmentation, Edge Detection, OpenCV, Object Detection, Model Benchmarking & Architecture Search

Deep Learning & ML: PyTorch, TensorFlow/Keras, Scikit-learn, NLP, Recommendation Systems, Reinforcement Learning

LLM & Generative AI: LangChain, RAG Systems, LLM Fine-tuning (LoRA, QLoRA, PEFT), Prompt Engineering, Chatbot Development, Vector Databases (FAISS, Chroma, Pinecone)

Data Science: NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering, EDA, Time Series Analysis

ML Systems & Infrastructure: ML Pipelines, Model Evaluation, Hyperparameter Tuning, Model Deployment, Experiment Tracking

Distributed Systems: Hadoop Ecosystem, HDFS, MapReduce, Distributed Data Processing

DevOps & Tools: Git, GitHub, Docker, Linux, REST APIs

Training

Computer Vision Trainee — Fawry *2026 – Present*

- Currently undergoing hands-on Computer Vision training at Fawry, building practical experience with real-world vision systems and production-oriented workflows.
- Applying deep learning techniques to image-based problems, reinforcing skills in model design, evaluation, and deployment within an industry setting.

Machine Learning Engineer Trainee — Digital Egypt Projects Initiative (DEPI) *Mar 2023 – 2024*

Ministry of Communications and Information Technology (MCIT)

- Developed deep learning architectures for Computer Vision and NLP applications using PyTorch.
- Conducted model experimentation, optimization, and performance evaluation across diverse image and text datasets.
- Explored scalable ML systems design and distributed data processing pipelines.

Executive Director & Machine Learning Instructor — PIE AI Student Activity *Ongoing*

- Led and managed the Machine Learning track, overseeing curriculum development and student mentorship.
- Designed and delivered ML and Computer Vision sessions aligned with DeepLearning.AI global curriculum standards.
- Mentored students on AI/ML fundamentals, project building, and industry best practices.

COMPUTER VISION PROJECTS

Agriculture Africa Infrastructure with satellite .

- Collecting data from satellite_02 and make some analysis on it to understand the land state and send the message to the user in WhatsApp about its land make a forecasting to the data every 3 days.
- I work with this idea on STISA competition.

Recycle Net — *Computer Vision | PyTorch | CNN*

- Designed and trained 18 custom Computer Vision models for multi-class image classification.

- Conducted systematic benchmarking and architecture search to identify the best-performing model.

DIP-GUI-Toolkit — *Python | Tkinter | Image Processing*

- Developed an interactive image processing toolkit with filtering, edge detection, and segmentation capabilities.
- Supported both grayscale and RGB image operations with a responsive GUI.

Economic Forecasting System — *XGBoost | LSTM | Time Series Analysis*

- Built forecasting models for GDP, inflation, and unemployment using LSTM and gradient boosting.
- Applied advanced feature engineering and time-series preprocessing techniques.

EDUCATION

Bachelor of Artificial Intelligence — Menoufia University *2023 – 2027 (Expected)*

Specialization: Machine Intelligence

Advanced Self-Study

- Designing Data-Intensive Applications — Martin Kleppmann
- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow — Aurélien Géron
- Computer vision application and algorithms- Richard Szeliski
- Deep Learning — Ian Goodfellow

CERTIFICATIONS

- AI Engineering Bootcamp: From Beginner to Expert — Udemy (2025)
- Diploma in Machine Learning — Udemy (2024)

LANGUAGES

Arabic (Native) • English (B2) • French (A2)